MOHAMMED SUHAIL

Aspiring VLSI / Semiconductor Engineer | Embedded Systems & Robotics

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OBJECTIVE

Second-year Electronics and Communication Engineering student with hands-on experience in RTL simulation (Verilog/VHDL, Xilinx Vivado), analog circuit simulation (KiCad, SPICE), and ESP32 firmware in C/C++. Seeking a VLSI / Semiconductor Engineering internship to apply digital design fundamentals and learn industry design flows, with a strong interest in embedded systems and robotics.

EDUCATION

Jyothy Institute of Technology — Bangalore, Karnataka
Bachelor of Engineering in Electronics and Communication Engineering (currently in 2nd year) — *Aug 2025 – Jun 2029*
Affiliated to Visvesvaraya Technological University

Deeksha Jnana Sweekar College — Bangalore, Karnataka
Pre-University Course — *Jan 2023 – Mar 2025*

TECHNICAL SKILLS

HDLs: Verilog, SystemVerilog, VHDL

Digital Design: Adders, counters, registers, ALU, memory, serial protocols (UART, SPI, I2C)

EDA & Simulation: Xilinx Vivado, KiCad (circuit design, SPICE simulation, PCB layout), Tinkercad, Wokwi

Programming: C, Embedded C, C++, Python

Hardware & Microcontrollers: ESP32, Arduino Nano, Arduino WROOM, servomotors, ultrasonic and medical/biometric sensors

Lab Skills: Breadboarding, soldering, oscilloscope, multimeter, logic analyser

Other: JavaScript, CSS, React, Node.js

PROJECTS

ALU Design and Verification | *Verilog, Xilinx Vivado*

- Designed and simulated an Arithmetic Logic Unit (ALU), the processor block that performs arithmetic and logic operations, in Verilog using Xilinx Vivado; verified its functionality using a testbench and simulation waveforms

Aether-Bot: Autonomous Medical Assistant Robot | *ESP32, C/C++, Servomotors, Ultrasonic Sensors*

- Designed and built an ESP32-based medical assistant robot in a team of 2–3, with multi-sensor integration for patient-care environments
- Integrated ultrasonic sensors for autonomous obstacle avoidance and a dual-servo mechanical arm for physical tasks
- Developed C/C++ firmware to read multiple sensors at the same time and drive synchronized LED status indicators

Analog Circuit Design and SPICE Simulation | *KiCad, SPICE*

- Designed and simulated amplifier (signal boosting), filter (frequency selection), and rectifier (AC-to-DC conversion) circuits using SPICE in KiCad

RouteVelo: Smart Transit Tracking Platform | *Product Requirements, System Architecture*

- Authored the Product Requirements Document and system architecture for a public transit tracking platform for Karnataka's transit network
- Conducted market research and SWOT analysis to define user personas, feature roadmap, and scalability constraints

CERTIFICATIONS

Building Multi-Agent Systems with Strands Agents – Amazon Web Services (AWS) — Mar 2026
AWS Machine Learning Learning Plan – Amazon Web Services (AWS) — Feb 2026
Machine Translation in Language Learning and Teaching – The Open University — Feb 2026

ACTIVITIES & INTERESTS

- Participated in a college-level hackathon

Spoken Languages: English, Hindi, Kannada

Interests: Embedded systems, medical robotics, circuit design and simulation